



# SQL CLASS NOTE – MODULE 7: LIMIT

## 1. Introduction

The **LIMIT clause** is used to control **how many rows** are returned by a query.

- Without **LIMIT** : SQL returns **all matching rows**.
- With **LIMIT** : SQL returns **only a specified number of rows**.

👉 Think of it as telling SQL:

- “Don’t show me everything, just give me the first X results.”

## 2. Syntax

```
SELECT column1, column2, ...  
FROM table_name  
LIMIT number;
```

- **number** = maximum rows to display.

You can also use **OFFSET** with **LIMIT** :

```
SELECT column1, column2  
FROM table_name  
LIMIT number OFFSET start;
```

- **OFFSET** = how many rows to skip before starting to display results.

## 3. Practical Examples with SchoolDB

### Example 1: Show the first 5 students

```
SELECT FirstName, LastName, Age
FROM Students
LIMIT 5;
```

### Example 2: Show the first 10 enrollments

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
LIMIT 10;
```

### Example 3: Show only 3 unique grade levels

```
SELECT DISTINCT GradeLevel
FROM Students
LIMIT 3;
```

### Example 4: Use OFFSET – skip the first 5 students, then show the next 5

```
SELECT FirstName, LastName, Age
FROM Students
LIMIT 5 OFFSET 5;
```

📌 Useful for **pagination** (breaking results into pages).

## Example 5: Show the 3 most recent enrollments (combine ORDER BY + LIMIT )

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
ORDER BY EnrollmentDate DESC
LIMIT 3;
```

## 4. Why Use LIMIT ?

- Makes queries **faster** by returning fewer rows.
- Useful when testing queries (don't want to see thousands of rows).
- Needed for **pagination** in apps (page 1 = first 10 rows, page 2 = next 10, etc.).

## 5. Real-Life Use Cases

- **Schools:** Show only the top 10 students in an exam ranking.
- **Banks:** Show a customer's last 5 transactions.
- **Hospitals:** Display the latest 3 patient admissions.
- **E-commerce:** Show only the first 20 products in search results.

## 6. Classwork (Practice)

👉 Try these queries on **SchoolDB**:

1. Display the first 7 students.
2. Show the first 5 courses.
3. Display only 3 unique teachers.
4. Show the first 10 enrollments ordered by EnrollmentDate ASC .
5. Use OFFSET to skip the first 10 students, then display the next 5.
6. Show the top 5 oldest students.
7. Show the latest 5 enrollments.

## 7. Challenge Question

👉 Write a query to display the **top 3 youngest female students**, showing their `FullName` , `Age` , and `GradeLevel` .

✅ That's the **expanded Module 7: LIMIT**.

Perfect 👍 Let's solve the **Module 7 Classwork + Challenge** step by step with real SQL queries.



## Solutions to Module 7: LIMIT

### 1. Display the first 7 students.

```
SELECT FirstName, LastName, Age, GradeLevel
FROM Students
LIMIT 7;
```

### 2. Show the first 5 courses.

```
SELECT CourseName, CourseCode, Teacher
FROM Courses
LIMIT 5;
```

### 3. Display only 3 unique teachers.

```
SELECT DISTINCT Teacher
FROM Courses
LIMIT 3;
```

#### 4. Show the first 10 enrollments ordered by EnrollmentDate ASC .

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
ORDER BY EnrollmentDate ASC
LIMIT 10;
```

#### 5. Use OFFSET to skip the first 10 students, then display the next 5.

```
SELECT FirstName, LastName, Age
FROM Students
LIMIT 5 OFFSET 10;
```

#### 6. Show the top 5 oldest students.

```
SELECT FirstName, LastName, Age
FROM Students
ORDER BY Age DESC
LIMIT 5;
```

#### 7. Show the latest 5 enrollments.

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
ORDER BY EnrollmentDate DESC
LIMIT 5;
```



## Challenge Question

👉 Display the **top 3 youngest female students**, showing their FullName , Age , and GradeLevel .

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName,  
        Age,  
        GradeLevel  
FROM Students  
WHERE Gender = 'Female'  
ORDER BY Age ASC  
LIMIT 3;
```

📌 This gives the **3 youngest female students** in the database.

✅ That completes **Module 7: LIMIT**.

👉 Should I move on to **Module 8: BETWEEN** in the same detailed style (definitions, examples, classwork, and challenge)?